

PAPER_897: Phonon Modulated Energy E_net Phonon

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Abstract

Phonon-modulated energy $E_{\text{net}}^{\text{phonon}}(t) = E_{\text{net}}(t) \times \Phi_{\{1.25\text{THz}\}}(\omega)$ with symmetric phonon pairing verification: $E_{\text{phonon}}^{+} + E_{\text{phonon}}^{-} = E_{\text{net}}^{\text{phonon}}$ (identity verified). Full sweep across buoyancy ratios showing regime transitions.

1. Core Equations

$$E_{\text{net}}^{\text{phonon}} = E_{\text{net}} \times \Phi$$
$$E_{\text{phonon}}^{+} + E_{\text{phonon}}^{-} \equiv E_{\text{net}}^{\text{phonon}}$$

2. Parameters

Parameter	Default	Description
E_0	1.0 J	Initial energy scale
t	0.0 s	Time parameter
F_UBi_over_FU	0.8	Buoyancy-to-field ratio
omega	$2\pi \times 1.25\text{e}12$ rad/s	Angular frequency

3. UQFF Integration

This calculator integrates `et_phonon_resonance.py` from Session 208 into the CP4 pipeline as class #481.

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